

NOUNCE: PERCEIVED-LABEL FINE-TUNING FOR ENGLISH PRONUNCIATION ASSESSMENT

“detecting what learners say — not what the dictionary says”

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1. Problem & the Data Gap

Turkish-native learners lack accessible, specific pronunciation feedback:

- **Limited access** to native speakers / quality instruction
- **Generic feedback** — “try again,” no phonetic detail

The deeper obstacle is technical: phonetic models like **POWSM** are trained on **canonical** (dictionary) transcriptions — produced by *automatic* IPA methods — so they *normalize away* the very errors a tutor must surface.

The data gap: almost no public corpus gives the **perceived** (produced) phone sequence for L2 English — the one signal a deviation detector must learn from. We bridge it with **L2-ARCTIC** (canonical + perceived on identical audio) and a small expert-annotated Turkish set.

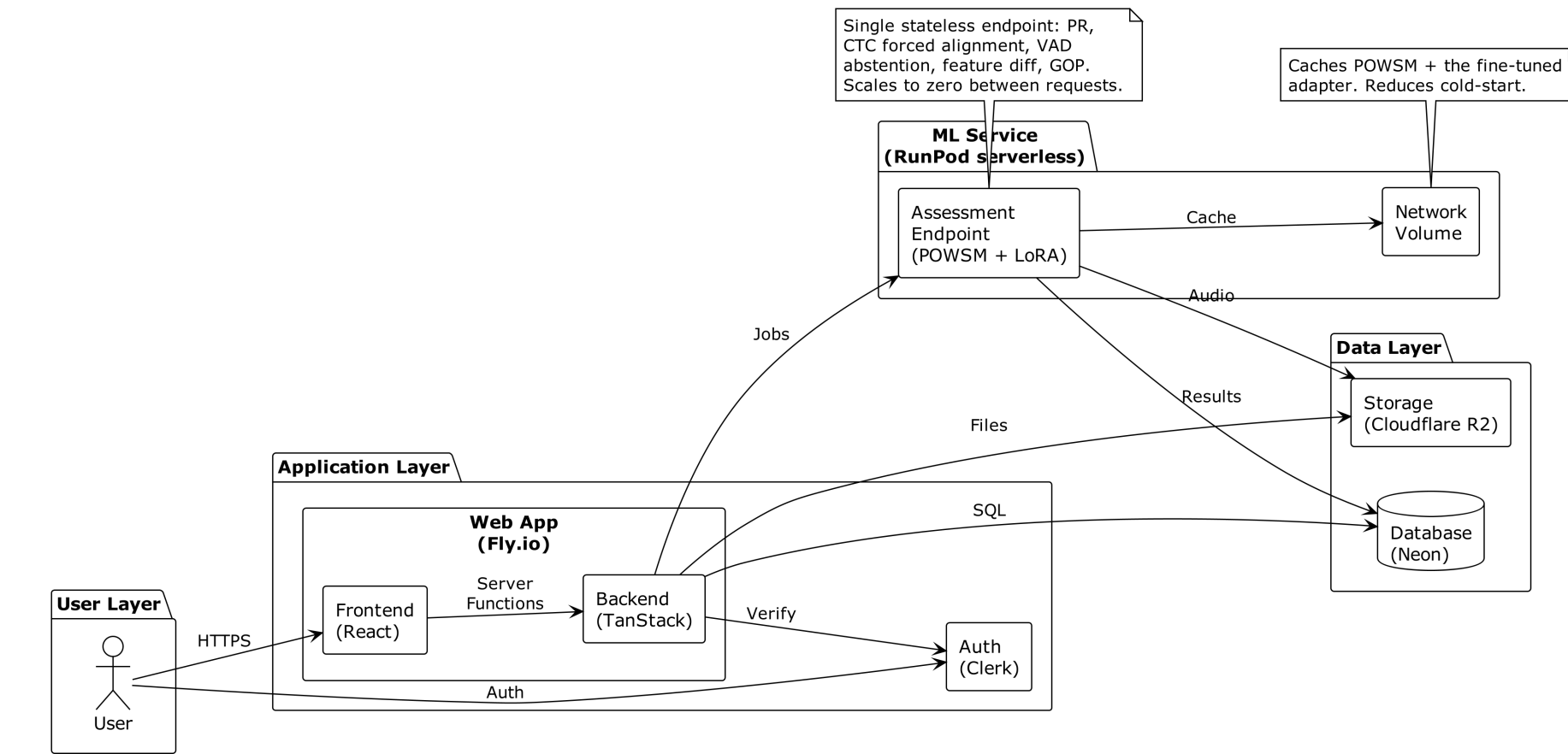
Canonical vs. Perceived

[GRAPHIC: canonical vs. perceived]

one /wʌn/ (canonical)
→ [wʌn] [wʌŋ] [wʌŋ] (perceived)
differing phones highlighted in red/blue

2. System (V1 → V2)

A reliable, deviation-aware pipeline. Browser → TanStack app → a **single** stateless RunPod serverless POWSM worker → per-phone feedback; Neon + R2 below.



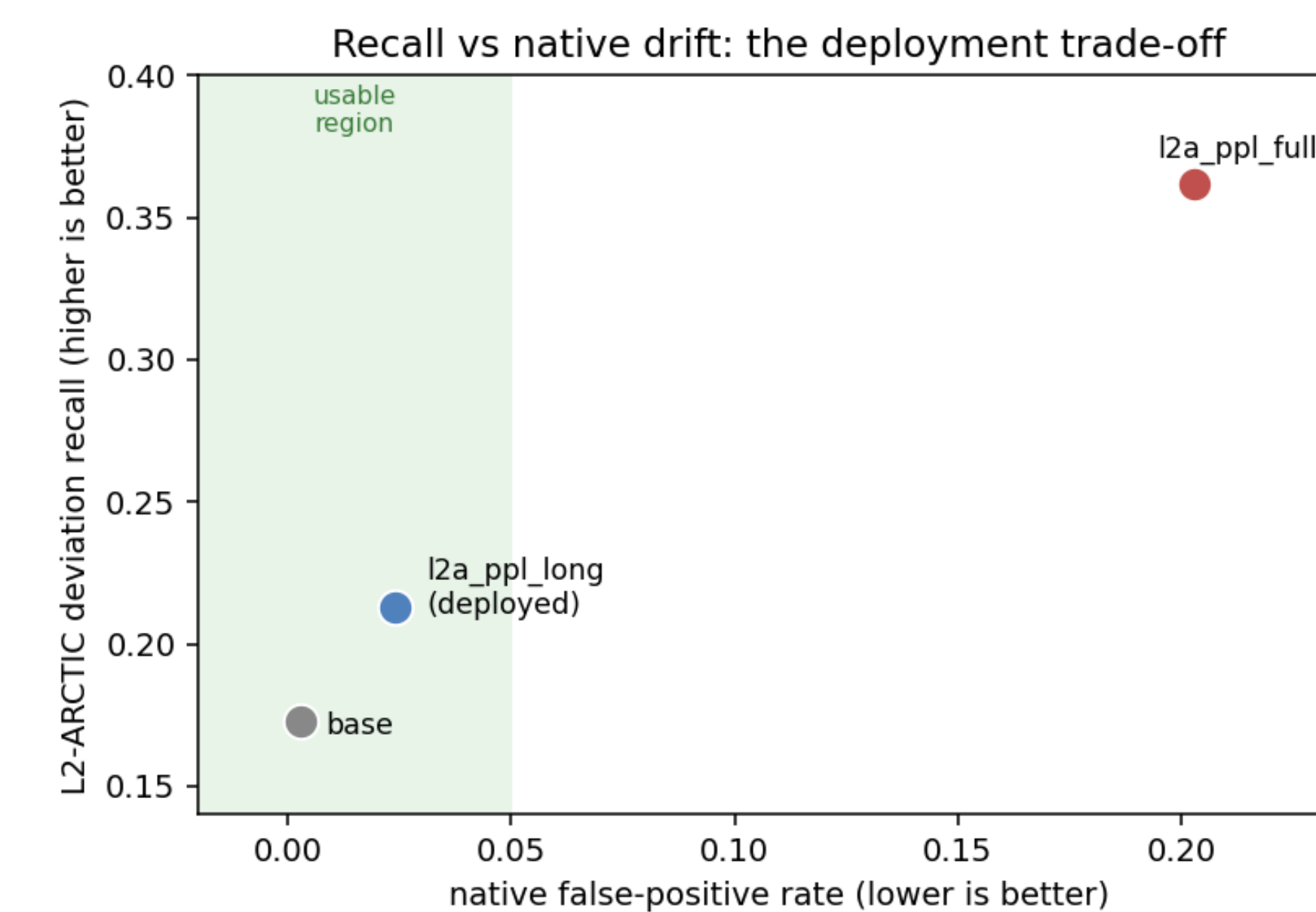
What changed since the prototype

V1 prototype	V2
MFA for timestamps	POWSM CTC forced alignment (one model, one inventory)
Two endpoints (G2P + PR)	One stateless worker
Exact-match phone diff	PanPhon articulatory feature distance
Scores any audio	Abstains on no-speech / noise / wrong sentence (VAD + SNR)
Off-the-shelf POWSM	+ Turkish-deviation-aware LoRA adapter (deployed)

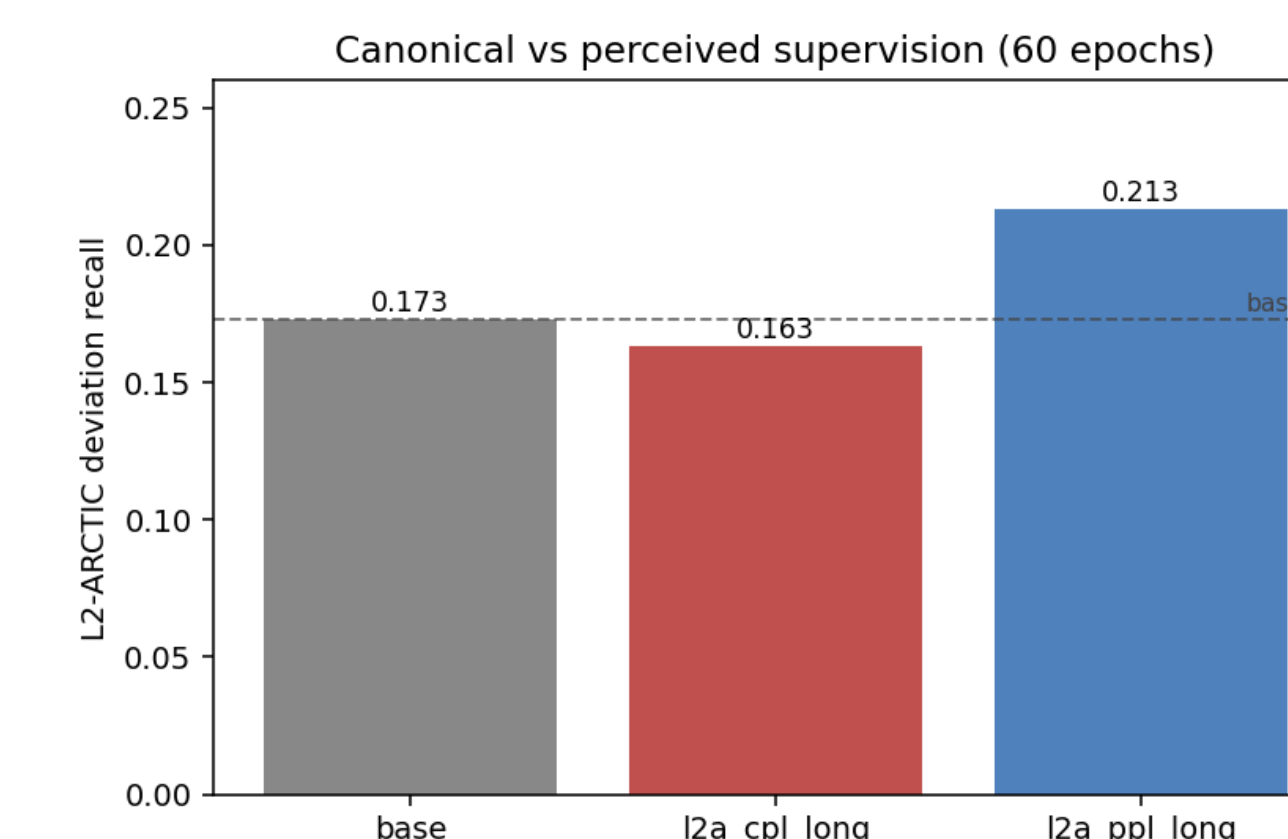
3. The Finding

A **controlled ablation**: same audio, same LoRA recipe ($r=32, \alpha=64$) — only the **label target** changes.

- **Canonical** → recall drops *below* baseline (**0.163** < **0.173**): *actively harms*.
- **Perceived** → recall **0.213** + best Turkish PER (**0.392**).
- Gap **widens** $\sim 4\times$ from 30 to 60 epochs.



Highest recall $\neq$ deployed: **l2a_pp1_full** flags 1-in-5 correct native phones. We ship the conservative **l2a_pp1_long**.

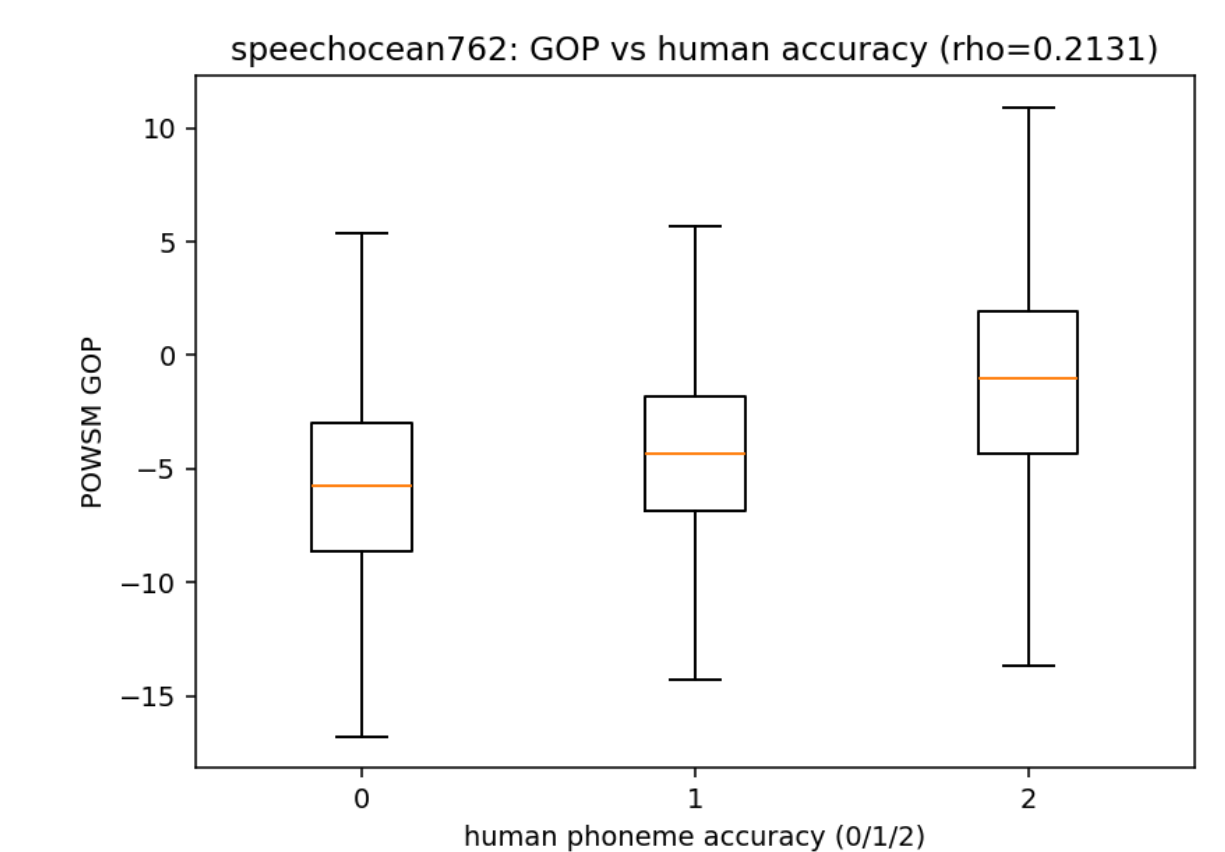


Convention beats quantity: the label target decides whether fine-tuning builds a *deviation detector* or just a better *transcriber*.

4. Validation

Human raters (blind, 0–10 intelligibility): system vs. mean rating Spearman $\rho = \mathbf{0.37}$ — *preliminary*: 2 raters, weak agreement (Krippendorff $\alpha = 0.11$); more raters is the bottleneck.

speechocean762 (47k phones, Mandarin-L1): GOP rises monotonically with expert accuracy; $\rho = 0.21$ phone-level, **0.37 sentence-level** — same 0.37 as the human study. Generalizes to a different L1.



Future Work

- **A perceived-annotated L2 corpus** — the single gating resource; with it, this recipe scales directly.
- **Per-accent adapters** — one small adapter per L1 over a frozen POWSM backbone.
- **Expert-in-the-loop data flywheel** — the app surfaces auto-annotations to linguists; corrections become training data → better adapters (TÜBİTAK panel, in progress).

github.com/umitcan07/senior | demo: **nounce.pro**
Turkish-native corpus: Öğr. Gör. Dr. Kardelen Kılıç, Eskişehir Tech. Univ.